

Chapter 1

Natural Community Types: Partitioning Natural Variation and Associating with Environmental Descriptors

1.1 Introduction

Before the contamination stress influence on the community can be examined in isolation, the influence of the natural environment must be taken into account. The community records of interest at any point in space and time are the joint outcome of natural environmental conditions, contamination pressure, and other, unrecorded sources of stress(Allan, 2004 [1]). Therefore a site's taxa record is a representation of all these factors acting together.

Two strategies are commonly used: adjusting for the natural environment within the model, or stratifying the data by it before the model is fitted. The first introduces the environmental variables as covariates in the regression of the community records of interest, so that contamination and natural conditions are modelled jointly and estimated finely; its drawback is that contamination gradients typically covary with natural environmental gradients (Graham, 2003 [2]), resulting collinearity with un-

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stable coefficients and limited practical value. The second strategy instead treats the natural environment as a categorical descriptor and partitions the sites into classes of similar environmental and(or) regional—characteristics (Hughes, Larsen, and Omernik, 1986 [3]; Allan, 2004 [1]) to avoid introducing covariates. This stratification can be a more interpretable approach, since it isolates the contamination signal within definable and comparable habitat stratas and divides the community records of interest with natural variation into community types with aligned environmental conditions.

However, the difficulty is practical rather than conceptual. Natural community composition is shaped by a large set of environmental features, many of which are unobservable, observable but not recorded, or recorded only with substantial measurement uncertainty. Consequently, in most empirical settings, the limited set of measured environmental variables is insufficient to define site groups that are genuinely homogeneous in both environmental conditions and the community records of interest. This limitation is illustrated in Figure 1.1. Complete environmental observation aligns the partition in Ω_{env} with the true community boundary; incomplete observation yields only a degraded partition that mislabels the sites between it and the true boundary—an error of incomplete descriptors, not of a failed environment–community relationship.

For this reason, we reverse the order in which environment and community enter the stratification. We first let the community data *define* the types directly, and then ask what environmental signature *describes* them. It rests on the previous conclusion: within the least-contaminated set, the recorded assemblages retain natural community composition. Therefore we can read natural community types from these least-contaminated sites, since they carry natural variation that we wish to stratify by. Starting with the least-contaminated set, we cluster its community matrix, validate and retain K branches as natural community types. Next, we model this environment-assemblage association by training a classifier that maps their environmental descriptors onto the defined community types. Through this process, we uncanonically stratify community types on the taxa space Ω_{taxa} , makes the implicit environment-determination assumption become an explicit and measurable property in the classifier building. This *define-then-describe* workflow is illustrated in Fig-

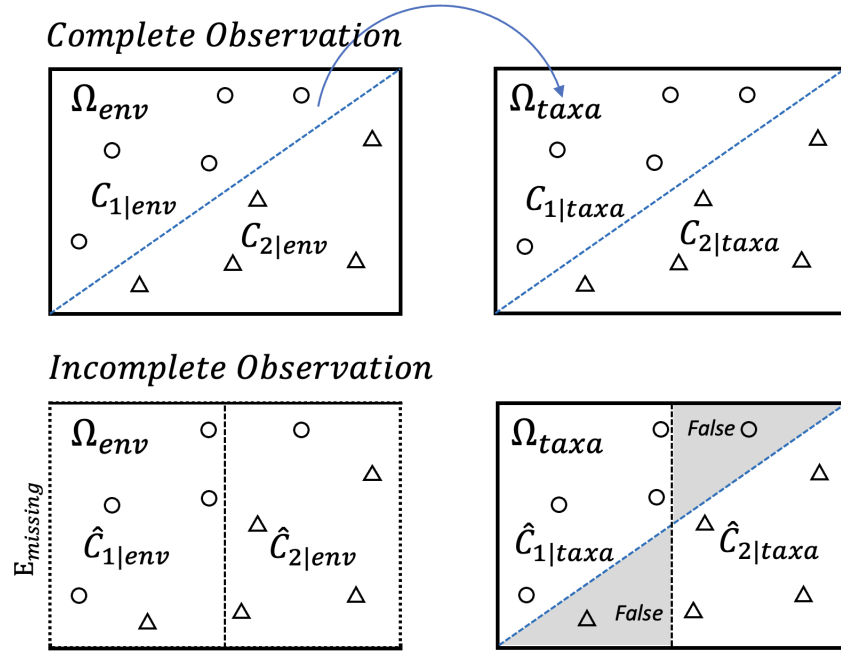


Figure 1.1: Under the assumption that environment determines community, incomplete environmental observation propagates classification error from Ω_{env} to Ω_{taxa} . Complete observation recovers the true diagonal boundary; incomplete observation imposes a degraded vertical partition, mislabelling sites in the wedge between the two (shaded, “False”).

ure 1.2.

The built classifier is then applied to the remaining, contaminated sites: predicting their environmental descriptors into community types. Through this process, every site regardless of contamination status is assigned to its natural community type in a broad sense.

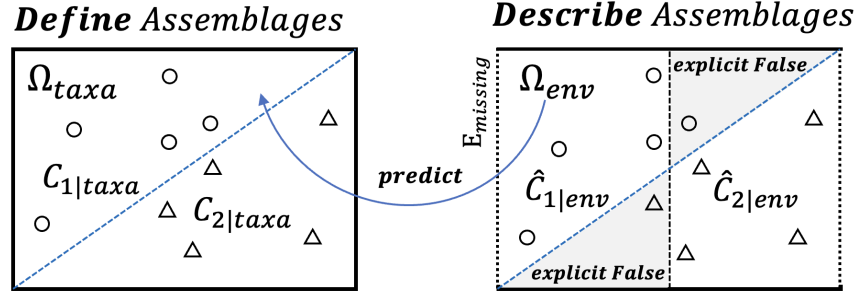


Figure 1.2: Community types are *defined* on the taxa space Ω_{taxa} , then *described* and predicted from the environmental space Ω_{env} . Limited environmental features make some misassignments, but these are rendered explicit (shaded, “explicit False”).

1.2 Methods

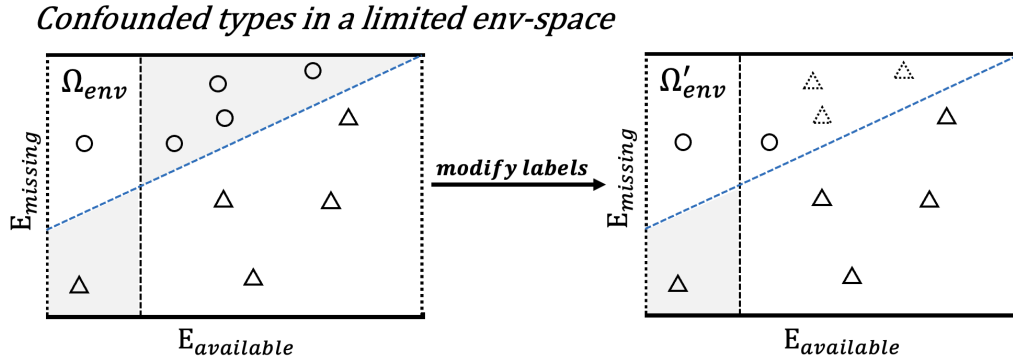


Figure 1.3: Confounded community types in a limited environmental space. The true boundary (blue dashed) depends on an unobserved feature $E_{missing}$, leaving the types overlapping (shaded) on the available features $E_{available}$. The original approach relabels part of C_1 as C_2 in one direction only ($\Omega_{env} \rightarrow \Omega'_{env}$, dotted triangles), reducing but not removing the overlap—separability bought by altering true labels.

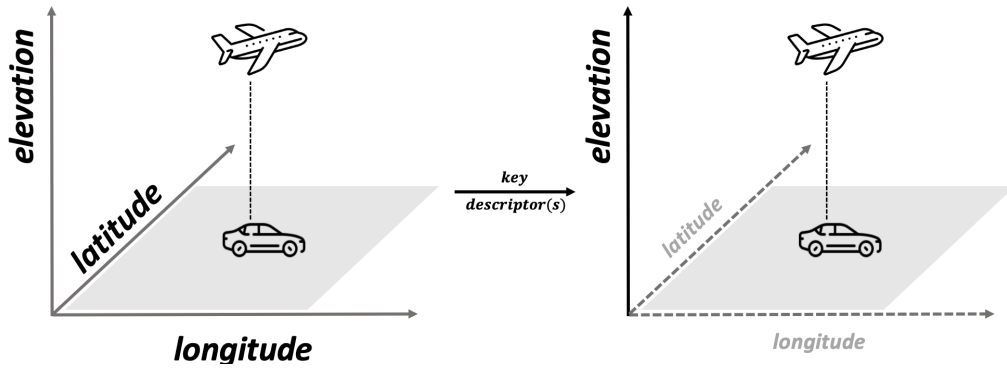


Figure 1.4: Not all descriptors help separate a small number of classes. An aircraft and a car are barely distinguishable on the horizontal descriptors (longitude, latitude), yet a single key descriptor—elevation—separates them cleanly. When only a few classes must be told apart, the right descriptor subset matters more than the number of descriptors, and retaining irrelevant ones only adds noise.

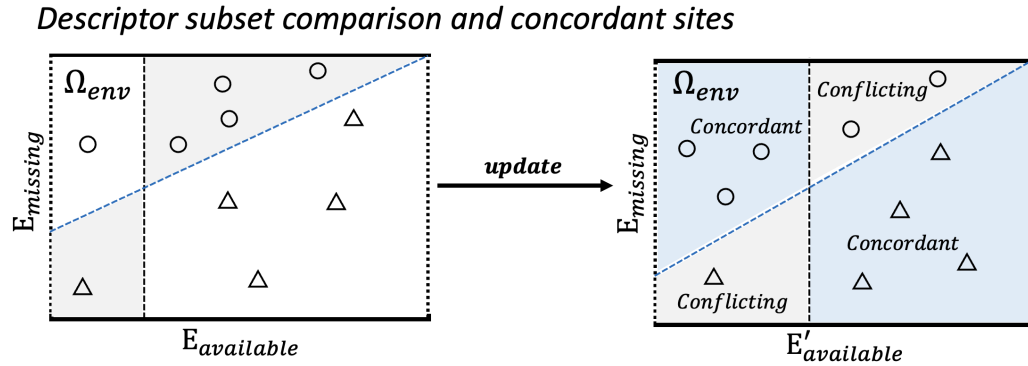


Figure 1.5: Selecting a descriptor subset and identifying concordant sites. On the initial descriptors $E_{available}$, the taxa-defined types overlap (left). Choosing a more discriminating subset $E'_{available}$ (right) separates most sites into *concordant* regions (blue), where taxa membership and environmental position agree, while the residual *conflicting* regions (grey) mark sites whose two views disagree. Only concordant sites are used to train the classifier; conflicting sites are retained as diagnostics. In practice, concordance is resolved across four categories (concordant, taxa-only, env-only, ambiguous), here simplified to two.

References

- [1] J. David Allan. “Landscapes and Riverscapes: The Influence of Land Use on Stream Ecosystems”. In: *Annual Review of Ecology, Evolution, and Systematics* 35 (2004), pp. 257–284. DOI: 10.1146/annurev.ecolsys.35.120202.110122.
- [2] Michael H. Graham. “Confronting Multicollinearity in Ecological Multiple Regression”. In: *Ecology* 84.11 (2003), pp. 2809–2815. DOI: 10.1890/02-3114.
- [3] Robert M. Hughes, David P. Larsen, and James M. Omernik. “Regional Reference Sites: A Method for Assessing Stream Potentials”. In: *Environmental Management* 10.5 (1986), pp. 629–635. DOI: 10.1007/BF01866767.